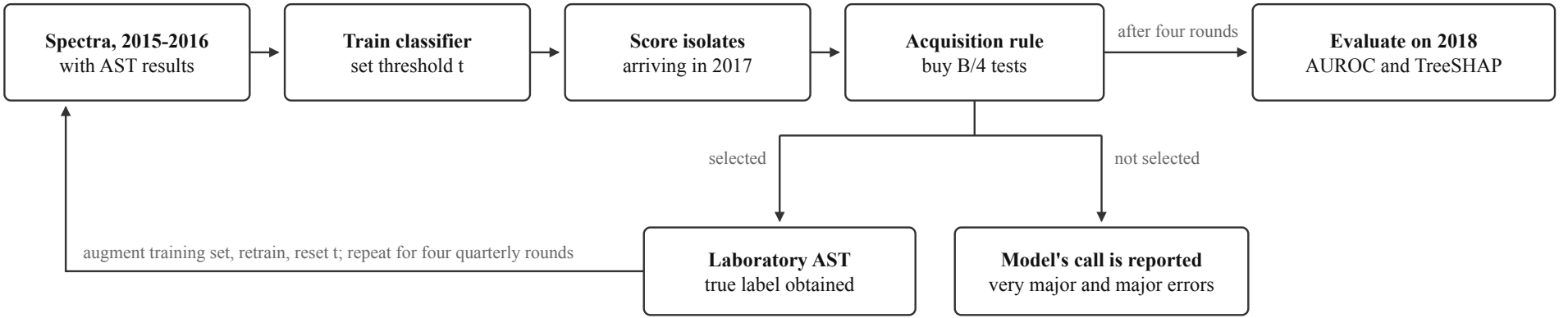



**Fig. 1.** Study pipeline. A classifier is trained on isolates from 2015 and 2016 and a decision threshold is fixed on held out data. Isolates arriving in 2017 are scored in four quarterly rounds. AST denotes antimicrobial susceptibility testing. In each round an acquisition rule buys laboratory susceptibility tests for a fixed share of the budget; those isolates return a true label and enter the training set, while the remaining isolates are reported using the model's own call and contribute to the error counts. The maintained model is evaluated on isolates from 2018.